

1 The actual third transition and its revived second parent

The released transition is the full physical interval $[t_3^a, t_3^a + \Gamma_3^{-1}]$, with $\dot{\alpha} = (1 - \eta_b)\mathcal{F}_2 + \eta_b\mathcal{F}_3$ and $\eta_b = \eta(\Gamma_3(t - t_3^a))$; see source (3.4)–(3.6), Section 3.7.5, pp. 30–31, and (5.22), p. 46. The following calculation retains that control, both evolving parents, their physical diffusion, and the original laboratory gravity. It does not use the exceptional fixed geometry of transition two. All older activation weights are one on this interval.

1.1 Exact determinant coordinates, coefficients, and inverse

Keep $A_0, \mu, \nu = \sqrt{A_0}$, $K_j = \mu\hat{\lambda}_j$, s_* , and the actual inherited scales σ_1, σ_2 . Physical viscosity and thermal diffusivity equal $d > 0$. Write

$$a = \sin s_2, \quad c = A_0 \cos s_*, \quad C = A_0 \sin s_*, \quad s = s_3 = L_3 \sigma_1 / \sigma_2, \quad \Gamma = \Gamma_3 = \sigma_2 \sin s. \quad (1)$$

Thus a, s and d are distinct parameters. Let $t_b = t_2^b$, let $r_b = |\zeta_2(t_b)|$, and put $b = r_b \sin s > 0$. The original third datum is $\zeta_3(t_b) = e(s)$ and $M_3(t_b) = I$, while $\zeta_2(t_b) = r_b e_2$. For $e(\phi) = (\sin \phi, \cos \phi)$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the two adjacent oriented determinants are exactly

$$\det(\zeta_1, \zeta_2) = -a, \quad \det(\zeta_2, \zeta_3) = -b. \quad (2)$$

Both identities persist through growth, transition and later controls whenever their equations exist. Indeed $|\zeta_1| = 1$ and ζ_1 also solves the transport equation of ζ_2 : the added first shear has transpose annihilating ζ_1 . Likewise the added second shear has transpose annihilating ζ_2 , so ζ_2, ζ_3 solve the same third transport equation. The determinant of two solutions of $\dot{z} = -D^T z$ has derivative $-\text{tr}(D)$ times itself, by differentiating its two columns. Every parent matrix has zero trace, proving both assertions.

Use the following coordinates, with their original dimensional factors:

$$\begin{aligned} \zeta_2 &= h\zeta_1 - aJ\zeta_1, & \zeta_3 &= x\zeta_1 - yJ\zeta_1, & D &= h^2 + a^2, \\ g &= \zeta_2 \cdot \zeta_3 = hx + ay, & hy - ax &= b, & x &= \frac{hg - ab}{D}, & y &= \frac{ag + hb}{D}, \\ E_1 &= -K_1\Theta_1 = A_1, & E_2 &= -K_2\Theta_2 = A_2/\sqrt{D}, & R &= |\zeta_3|^2 = \frac{g^2 + b^2}{D}. \end{aligned} \quad (3)$$

Here D and R are scalars, not velocity-gradient matrices. In particular the actual second amplitude is $A_2 = \sqrt{D}E_2$; replacing it by E_2 would lose the evolving phase length. Put $\theta = \phi_2$, so the full phase and angular inverse is

$$\begin{aligned} \zeta_1 &= \frac{1}{\sqrt{D}}(h \sin \theta - a \cos \theta, h \cos \theta + a \sin \theta), \\ \zeta_2 &= \sqrt{D}(\sin \theta, \cos \theta), \\ \zeta_3 &= \frac{1}{\sqrt{D}}(g \sin \theta + b \cos \theta, g \cos \theta - b \sin \theta), \\ \phi_1 &= \theta - \arctan(a/h), & \alpha &= s_* - \phi_1. \end{aligned} \quad (4)$$

The domain used below has $h, g > 0$ and $g \cos \theta - b \sin \theta > 0$; hence these expressions select the positive second-component branches. The original deformation matrices are

$$M_1 = R_\alpha, \quad M_2 = R_{\alpha-s_*} \begin{pmatrix} 1 & -(h - \cos s_2)/a \\ 0 & 1 \end{pmatrix}, \quad M_3 = \mathcal{L}(t)^{-T} \mathcal{L}(t_b)^T, \quad \mathcal{L} = (\zeta_2 \ \zeta_3). \quad (5)$$

The first two preserve their original activation data, and $\det \mathcal{L} = -b \neq 0$. Differentiating the last formula using $\dot{\mathcal{L}} = -D_{<3}^T \mathcal{L}$ gives $\dot{M}_3 = D_{<3} M_3$, $M_3(t_b) = I$, and $M_3^{-T} e(s) = \zeta_3$. All three have determinant one. Thus the inverse gives the complete original matrix evolution, rather than merely the newest phase direction.

Define the exact temperature numerators

$$N_2 = a(E_1 + c) + Ch, \quad N_3 = (E_1 + c)y + Cx + bE_2. \quad (6)$$

The full older physical equations are

$$\begin{aligned} \dot{h} &= a\Omega_1, & \dot{g} &= \Omega_1 \frac{2ahg + (h^2 - a^2)b}{D} + b\Omega_2, \\ \dot{E}_1 &= -C\Omega_1 - dK_1^2 E_1, & \dot{\Omega}_1 &= -E_1 \zeta_{1,1} - dK_1^2 \Omega_1, \\ \dot{E}_2 &= -\frac{N_2}{D} \Omega_2 - dK_2^2 D E_2, & \dot{\Omega}_2 &= -E_2 \sqrt{D} \sin \theta - dK_2^2 D \Omega_2, \\ \dot{\theta} &= -\dot{\alpha} - \frac{a^2 \Omega_1}{D}. \end{aligned} \quad (7)$$

In terms of the actual amplitude A_2 the fifth equation is exactly

$$\dot{A}_2 = \left(\frac{ah\Omega_1}{D} - dK_2^2 D \right) A_2 - \frac{N_2}{\sqrt{D}} \Omega_2.$$

The exact source feedbacks and their factored difference are

$$\begin{aligned} \mathcal{F}_2 &= \frac{a\Omega_1(h \tan \theta - a)}{D}, \\ \mathcal{F}_3 &= \frac{\Omega_1 y(h \sin \theta - a \cos \theta) + b\Omega_2 \sin \theta}{g \cos \theta - b \sin \theta}, \\ \mathcal{F}_3 - \mathcal{F}_2 &= \frac{b\Omega_1(h \sin \theta - a \cos \theta)(h \cos \theta + a \sin \theta)}{D \cos \theta (g \cos \theta - b \sin \theta)} + \frac{b\Omega_2 \sin \theta}{g \cos \theta - b \sin \theta}. \end{aligned} \quad (8)$$

With η_b as specified, the final equation in (??) becomes

$$\dot{\theta} = -\frac{ah\Omega_1}{D} \tan \theta - \eta_b(\mathcal{F}_3 - \mathcal{F}_2). \quad (9)$$

At the actual transition entry $\theta = \Omega_2 = 0$, $\mathcal{F}_3 - \mathcal{F}_2 = -abh\Omega_1/(Dg) < 0$. The switching therefore drives a positive second angle and a negative second vorticity, as proved on the full interval below.

Verification of the closed equations and exact invariants. The rotating base satisfies $R_\alpha e_2 = \cos s_* \zeta_1 + \sin s_* J \zeta_1$. Consequently

$$G_{<3} = -(E_1 + c + hE_2)\zeta_1 - (C - aE_2)J\zeta_1.$$

Taking $-J\zeta_2 \cdot G_{<2}$ and $-J\zeta_3 \cdot G_{<3}$ gives respectively N_2, N_3 ; the shear-amplitude coefficients are therefore $a_2 = N_2/(K_2 D)$ and $a_3 = N_3/(K_3 R)$, with their original signs. Since $\dot{\zeta}_1 = \dot{\alpha} J \zeta_1$, the phase equations in this moving orthonormal basis give, component by component,

$$\dot{h} = a\Omega_1, \quad \dot{x} = \Omega_1 y + \frac{bh}{D} \Omega_2, \quad \dot{y} = \frac{ba}{D} \Omega_2.$$

The derivative of $hy - ax$ is zero, and differentiating $g = hx + ay$ proves its displayed row. The damped amplitude rows then give every remaining row of (??). Substituting these phases in the source feedback formula proves (??); the polar phase equation gives (??). Conversely these calculations reverse: differentiating (??), with (??) and (??), gives the three physical phase equations and the source angular control. Together with (??), the temperature inverse and the vorticity rows, this proves equivalence on the stated domain.

There is an exact cancellation despite the revival of Ω_2 :

$$\dot{N}_2 = -adK_1^2 E_1, \quad \dot{N}_3 = -d(K_1^2 E_1 y + bK_2^2 D E_2). \quad (10)$$

Indeed

$$\begin{aligned}\dot{N}_3 = & (-C\Omega_1 - dK_1^2 E_1)y + (E_1 + c)\frac{ba}{D}\Omega_2 + C\left(\Omega_1 y + \frac{bh}{D}\Omega_2\right) \\ & + b\left(-\frac{a(E_1 + c) + Ch}{D}\Omega_2 - dK_2^2 DE_2\right),\end{aligned}$$

whose two shear groups cancel exactly. Differentiating N_2 gives its first formula. Thus both numerators are conserved when $d = 0$ even though the individual phases, amplitudes and controls evolve. For positive d the two losses in $\dot{N}_3$ are retained explicitly. $\square$

1.2 The general aggregate identity, including activation

The cancellation has an exact form for any finite active family. Write $B_j = w_j K_j \Theta_j \zeta_j$, $r_j = |\zeta_j|$, $E_j = -K_j \Theta_j$, and

$$G_{<q} = -A_0 R_\alpha e_2 + \sum_{j<q} B_j, \quad D_{<q} = \dot{\alpha} J + \sum_{j<q} w_j \Omega_j \frac{J \zeta_j \otimes \zeta_j}{r_j^2}.$$

Use the phase equations and the physical temperature rows with thermal diffusivity d , without added amplitude forcing. Direct pairwise cancellation gives

$$\dot{G}_{<q} + D_{<q}^T G_{<q} = \sum_{j<q} (\dot{w}_j K_j \Theta_j - dw_j K_j^3 r_j^2 \Theta_j) \zeta_j. \quad (11)$$

For precision, the j th shear applied to all lower-index gradient terms is $w_j \Omega_j \zeta_j (J \zeta_j \cdot G_{<j}) / r_j^2$. It cancels $w_j K_j \dot{\Theta}_j \zeta_j$ apart from diffusion. The terms with lower shears cancel the phase derivative in B_j , its self-shear vanishes because $J \zeta_j \cdot \zeta_j = 0$, and the rotational base and phase terms cancel exactly. The only terms left are precisely those displayed in (??), including $\dot{w}_j$.

Put $N_q = -J \zeta_q \cdot G_{<q}$ and $\Delta_{jq} = \det(\zeta_j, \zeta_q)$. For any trace-free 2×2 matrix D , direct multiplication gives $JD^T + DJ = 0$. Differentiating the bilinear pairing using $\dot{\zeta}_q = -D_{<q}^T \zeta_q$ thus gives

$$\dot{N}_q = \sum_{j<q} (\dot{w}_j - dw_j K_j^2 r_j^2) E_j (-\Delta_{jq}). \quad (12)$$

Equivalently its diffusion part is $\sum_{j<q} dK_j^2 r_j^2 J \zeta_q \cdot B_j$. The sign is fixed by $\det(\zeta_q, \zeta_j) = -\Delta_{jq}$, not by a convention about unoriented angle sizes. No nonadjacent determinant has been assumed constant. For $q = 3$, $-\Delta_{13} = y$, $-\Delta_{23} = b$, and $w_1 = w_2 = 1$; (??) gives exactly (??). During activation the term $\dot{w}_j E_j (-\Delta_{jq})$ remains, so inviscid constancy is asserted only after those weights become constant.

1.3 Full transition from the actual third-growth endpoint

Here are the actual endpoint bounds used in the finite proof. They are quantities at the already constructed threshold $t_a = t_3^a$, not assumptions about the subsequent unknown trajectory. The preceding third-growth calculation starts at the proved second return and gives

$$\begin{aligned}1 \leq r_b \leq \sqrt{10}, \quad h(t_a) \in [\sqrt{1-a^2}, 3), \quad 1 \leq D(t_a) < 10, \quad \theta(t_a) = \Omega_2(t_a) = 0, \\ E_1(t_a)/\sigma_1^2 \in [1/4, 2], \quad 0 < \Omega_1(t_a)/\sigma_1 < 9, \quad A_2(t_a)/\sigma_2^2 \in [2/5, 2], \\ 1 \leq R(t_a) \leq 2, \quad \frac{1}{2} \sin s \leq \zeta_{3,1}(t_a) \leq \sin s, \quad \zeta_{3,2}(t_a) \geq \cos s > 15/16, \\ \frac{2}{5} \leq \frac{N_3(t_a)}{\sigma_2^2 \sin s} \leq \frac{16}{15}, \quad \frac{1}{2} \leq \frac{\sigma_2 \Omega_3(t_a)}{K_3 \Theta_3(t_a)} \leq \sqrt{5}, \\ \Theta_3(t_a) = -\frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \quad 0 < c/\sigma_1^2 \leq 1/4, \quad 0 < C/\sigma_1^2 \leq a/32.\end{aligned} \quad (13)$$

In particular $E_2(t_a)/\sigma_2^2 \in [1/8, 2]$: the lower bound follows from $(2/5)/\sqrt{10} > 1/8$. Also $g(t_a) = \sqrt{D(t_a)}\zeta_{3,2}(t_a)$ lies in $[15/16, \sqrt{20}]$. These are the exact converted endpoint bounds needed below. Retain the explicit source restrictions

$$0 < a \leq 1/512, \quad 0 < s \leq a, \quad L_3 \geq 16384, \quad \delta_2 = \frac{dK_2^2}{\Gamma} \leq \frac{1}{128}, \quad \delta_3 = \frac{dK_3^2}{\Gamma} \leq \frac{1}{108}. \quad (14)$$

The preceding source-frequency selection may be strengthened by these explicit numerical lower bounds; it preserves the original recursion and scale rule. In physical units, $\sigma_1 \geq \sigma_{\min} > 0$ and $\Gamma \geq (15/16)L_3\sigma_1$ show that the nonempty range

$$0 < d \leq \min \left\{ \frac{15L_3\sigma_{\min}}{2048K_2^2}, \frac{L_3\sigma_{\min}}{128K_3^2} \right\}$$

satisfies the two diffusion restrictions. This is intersected with the preceding actual-growth range, whose entries are positive.

Theorem 1. *For the actual endpoint (??) and original parameters (??), the entire original third transition exists through $t_1 = t_a + \Gamma^{-1}$. Put $\tau = \Gamma(t - t_a)$, $u = \theta/s$, and $V = \Omega_2/\sigma_2$. Throughout $0 \leq \tau \leq 1$,*

$$0 \leq u \leq \frac{640a}{L_3}, \quad -\frac{7040a}{L_3} \leq V \leq 0. \quad (15)$$

Both inequalities have strict signs after the switching profile has become positive on an interval. The full older geometry obeys

$$\begin{aligned} \sqrt{1-a^2} \leq h < 13/4, \quad 1 \leq D < 11, \quad 3/4 < g < 5, \quad x, y > 0, \\ 1/8 < E_1/\sigma_1^2 \leq 2, \quad 0 < \Omega_1/\sigma_1 < 10, \quad 1/16 < E_2/\sigma_2^2 < 3, \\ \cos \theta > 15/16, \quad g \cos \theta - b \sin \theta > 2/3. \end{aligned} \quad (16)$$

All source feedback denominators are therefore positive, and (??) gives finite original matrices and inverses. The third numerator and phase length satisfy

$$e^{-11\delta_2\tau} N_3(t_a) \leq N_3(t) \leq N_3(t_a), \quad \left| \log \frac{|\zeta_3(t)|}{|\zeta_3(t_a)|} \right| \leq \frac{46000a}{L_3} < \frac{1}{128}. \quad (17)$$

The actual newest amplitude remains negative and its quotient $v = \sigma_2\Omega_3/(K_3\Theta_3)$ satisfies $1/2 \leq v \leq 3$. Its full transition logarithmic gain is bounded by

$$\frac{1}{18} \leq \log \frac{-\Theta_3(t_1)}{-\Theta_3(t_a)} \leq 4. \quad (18)$$

The exact exponent, including physical diffusion, is given in the proof. This theorem completes the transition; it does not assert a third shooting root or an infinite fixed-viscosity cascade.

Proof. Set $L = L_3$, $\epsilon = \sigma_1/\Gamma = s/(L \sin s) \leq 2/L$, $q = \sigma_1/\sigma_2 = s/L$, $E = E_1/\sigma_1^2$, $F = E_2/\sigma_2^2$, $\omega = \Omega_1/\sigma_1$, $\bar{c} = c/\sigma_1^2$, $\bar{C} = C/\sigma_1^2$ and $\delta_1 = dK_1^2/\Gamma \leq \delta_2$. Primes now mean $d/d\tau$ exclusively. The exact dimensionless older equations, retaining each physical ratio, are

$$\begin{aligned} h' &= a\epsilon\omega, & g' &= \epsilon\omega \frac{2ahg + (h^2 - a^2)b}{D} + \frac{b}{\sin s}V, \\ E' &= -\epsilon\bar{C}\omega - \delta_1 E, & \omega' &= -\epsilon E \frac{h \sin(su) - a \cos(su)}{\sqrt{D}} - \delta_1 \omega, \\ F' &= -q\epsilon \frac{a(E + \bar{c}) + \bar{C}h}{D} V - \delta_2 DF, & V' &= -F\sqrt{D} \frac{\sin(su)}{\sin s} - \delta_2 DV. \end{aligned} \quad (19)$$

The exact u' is (??) divided by Γs , with (??); no approximation of either feedback occurs.

Initially impose the temporary closed bounds $0 \leq u \leq 1/1024$, $-1/32 \leq V \leq 0$, $\sqrt{1-a^2} \leq h \leq 13/4$, $3/4 \leq g \leq 5$, $1/8 \leq E \leq 3$, $0 \leq \omega \leq 10$, and $1/16 \leq F \leq 3$. The sign boundaries are invariant: at $u = 0$, its vector field equals $\eta b \epsilon \omega a h / (s D g) \geq 0$; at $V = 0$ its vector field is nonpositive. Also $h \sin(su) < a \cos(su)$, since $s \leq a$ and $u \leq 1/1024$. The ω equation and its positive datum imply $\omega > 0$. These conclusions can equally be obtained from their integrating factors and a first-exit argument, so tangency of a sign boundary causes no loss of the sign assertion.

The temporary bounds give $D \geq 1$, because $h' \geq 0$ and $D(t_a) \geq 1$, and $D < 11$. They give $\cos(su) > 15/16$ and

$$g \cos(su) - b \sin(su) > \frac{3}{4} \frac{15}{16} - \frac{4a^2}{1024} > \frac{2}{3},$$

using $b \leq \sqrt{10} \sin s < 4s \leq 4a$. Thus every equation is smooth on a fixed neighborhood of this box. Because E decreases and $\omega' \leq \epsilon E a \leq 6a/L$,

$$E \leq 2, \quad 0 < \omega \leq 9 + 6a/L < 10, \quad h \leq 3 + 20a/L < 13/4.$$

Integrating the E row, using the temporary upper bounds, gives

$$E \geq \frac{1}{4} - \frac{20a}{32L} - \frac{3}{128} > \frac{1}{8}.$$

The numerator in the F equation obeys $0 < a(E + \bar{c}) + \bar{C}h < 4a$. Since $V \leq 0$, $F' \geq -11\delta_2 F$, hence $F \geq \frac{1}{8}e^{-11/128} > 1/16$. Its positive derivative contribution is at most $q\epsilon(4a)/32 \leq as/(4L^2)$, so $F < 3$. The older contribution to g' is positive, and

$$g \geq \frac{15}{16} - \frac{\sqrt{10}}{32} > \frac{3}{4}, \quad g \leq \sqrt{20} + \frac{300a}{L} < 5.$$

For the upper estimate use $[2ahg + (h^2 - a^2)b]/D \leq 2ag/h + b < 15a$ and $\epsilon\omega \leq 20/L$. Also $hg - ab > 0$ and $ag + hb > 0$, proving $x, y > 0$.

It remains to strictly improve the two revival bounds. On the box, the first term in (??) is nonpositive. In the factored feedback difference its first numerator has negative first factor; the other factor is positive. Its absolute forcing in u' is at most

$$\frac{2}{L} 4 \frac{10}{a} \frac{7}{2} \frac{16}{15} \frac{3}{2} = \frac{448a}{L} < \frac{512a}{L}.$$

Here $|h \sin \theta - a \cos \theta| \leq a$, $h \cos \theta + a \sin \theta < 7/2$, $b/s < 4$, and $D \geq 1$. The second-parent feedback term is at most $6|V|u$ because $b/\sin s < 4$, $\sin(su)/s \leq u$, and the second denominator exceeds $2/3$. Therefore

$$u' \leq 512a/L + (3/16)u, \quad u(\tau) \leq (512a/L)e^{3/16} < 640a/L < 1/1024.$$

The elementary estimate $e^{3/16} < 5/4$ follows, for example, from $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$. For $Z = -V \geq 0$, the integrating factor in its equation gives

$$Z(\tau) \leq \int_0^\tau F \sqrt{D} \frac{\sin(su)}{\sin s} dr < 11 \int_0^\tau u dr \leq 7040a/L < 1/32,$$

since $3\sqrt{11}(16/15) < 11$ and $s/\sin s < 16/15$. Every boundary has now been strictly improved, or has its invariant sign. The original positive data and integrating factors keep the positive quantities away from zero. The vector field remains in a compact subset of its smooth domain; its integral equation and the local existence theorem therefore extend it through $\tau = 1$. This proves the full prescribed interval, not only a small initial piece. Once η is positive on an interval, the strictly positive forcing at $u = 0$ gives $u > 0$ thereafter, and the integral for Z then gives $V < 0$. No instantaneous derivative at a flat endpoint is used to assert this revival.

For the numerator, $E_1, E_2, x, y, c, C > 0$ and $D \geq 1$ show

$$0 \leq K_1^2 E_1 y + b K_2^2 D E_2 \leq K_2^2 D N_3.$$

Equation (??) and its integrating factor give the first part of (??). For the length, the exact polar transport equation gives

$$|(\log |\zeta_3|)'| \leq \epsilon \omega |\sin(\phi_3 - \phi_1)| + \frac{|V|}{\sin s} |\sin(\phi_3 - \phi_2)|.$$

The determinant coordinates give

$$|\sin(\phi_3 - \phi_1)| = \frac{ag + hb}{\sqrt{D}\sqrt{g^2 + b^2}} \leq a + b/g < 7a, \quad |\sin(\phi_3 - \phi_2)| = \frac{b}{\sqrt{g^2 + b^2}} < 6s.$$

The cosine factors in the exact polar equation have absolute value at most one; these inequalities thus retain its full contribution. Consequently

$$|(\log |\zeta_3|)'| \leq 140a/L + (32/5)|V| \leq 45196a/L < 46000a/L < 1/128.$$

The last numerical inequality uses $a \leq 1/512$, $L \geq 16384$. This proves the second part of (??), and in particular $R(t) < 2e^{1/64} < 3$.

For the newest linear pair define the exact scaled coefficients

$$\mathcal{K} = \frac{N_3}{\sigma_2^2 \sin s R}, \quad \mathcal{B} = \frac{\zeta_{3,1}}{\sin s}.$$

At entry $1/5 \leq \mathcal{K}(t_a) \leq 16/15$. The proved variations give

$$\frac{1}{6} < \frac{1}{5}e^{-13/128} \leq \mathcal{K} \leq \frac{16}{15}e^{1/64} < \frac{9}{8}.$$

For the lower strict inequality use $e^{-13/128} \geq 115/128 > 5/6$; for the upper use $e^{1/64} \leq 64/63$. Furthermore $(\log \sqrt{D})' = a h \epsilon \omega / D \leq 22a/L < 1/64$. Because the second phase is vertical at entry,

$$\frac{\zeta_{3,1}(t)}{\zeta_{3,1}(t_a)} = \frac{\sqrt{D(t_a)}}{\sqrt{D}} \left(\cos \theta + \frac{g}{b} \sin \theta \right).$$

Its lower bound is $(63/64)^2$, and its upper bound is $1 + 6u \leq 1 + 3840a/L < 9/8$, using $b \geq \sin s$. Thus

$$1/6 < \mathcal{K} < 9/8, \quad 15/32 < \mathcal{B} < 9/8. \quad (20)$$

Construct the Riccati equation $v' = \mathcal{B} - \mathcal{K}v^2$ with its actual entry value. Its field points strictly inward at $v = 1/2$ and $v = 3$, since $15/32 - (9/8)/4 = 3/16 > 0$ and $9/8 - 9/6 = -3/8 < 0$. It exists on the entire interval with $1/2 \leq v \leq 3$. Then define

$$P_3(t) = P_3(t_a) \exp\left(\int_0^t (\mathcal{K}v - \delta_3 R) dr\right), \quad \Theta_3 = -P_3, \quad \Omega_3 = -\frac{K_3}{\sigma_2} P_3 v. \quad (21)$$

Differentiating these expressions gives exactly $\dot{\Theta}_3 = N_3 \Omega_3 / (K_3 R) - dK_3^2 R \Theta_3$ and $\dot{\Omega}_3 = K_3 \zeta_{3,1} \Theta_3 - dK_3^2 R \Omega_3$. The inverse is thus the unique physical linear pair, by its integral equation; it has not assumed that its temperature avoids zero. Finally $\mathcal{K}v \geq 1/12$, $\delta_3 R < 3/108$, and $\mathcal{K}v < 27/8 < 4$ give $1/18 \leq \mathcal{K}v - \delta_3 R \leq 4$. Integrating over the full unit transition proves (??). $\square$

The exact first, second and third diffusion exponents during this transition are respectively δ_1 , $\delta_2 \int_0^1 D d\tau$ and $\delta_3 \int_0^1 R d\tau$. They equal $dK_j^2 \int_{t_a}^{t_1} |\zeta_j(t)|^2 dt$ in physical coordinates. The second phase and amplitude continue evolving inside these integrals. The end control is exactly $\dot{\alpha} = \mathcal{F}_3$, so the source next steering starts smoothly, with its actual values $Z_3(t_1) = \zeta_{3,1}(t_1)$ and $|\zeta_3(t_1)|$. The released periodic profile is not a Laplacian eigenfunction; realizing these damped trajectories with its compact spatial fields still requires the explicitly calculated profile and localization forces. None of the determinant or transition results discards those forces or asserts an unforced spatial superposition.